Lesson 6.6 **Duration: 40 minutes**

Cloud Computing

Learning Objectives

- Define cloud computing and explain its service models (IaaS, PaaS, SaaS)
- Compare on-premise computing with cloud-based solutions
- Analyze the economic and operational benefits of cloud adoption

Engage (5-7 minutes)

Ask students where their files are stored when they use Google Drive, stream music on Spotify, or play online games. They'll say "the cloud." Then ask: what actually is the cloud? Reveal that it's simply someone else's computer—massive data centers with enormous computing power, available over the internet. This demystification sets up the technical discussion.

Explore (10-12 minutes)

Students compare two scenarios: a school running its own email server (buying hardware, hiring IT staff, maintaining backups) versus using Google Workspace (paying a monthly fee, automatic updates, no hardware). They calculate rough costs for each approach over 3 years, including hardware, electricity, maintenance, and personnel. Discuss when each approach makes sense.

Explain (10-12 minutes)

Cloud computing delivers computing services—servers, storage, databases, networking, software—over the internet on a pay-as-you-go basis. Three service models: Infrastructure as a Service (IaaS) provides virtual machines and storage (AWS EC2, Azure VMs); Platform as a Service (PaaS) provides development environments (Google App Engine, Heroku); Software as a Service (SaaS) delivers complete applications (Google Workspace, Microsoft 365). Deployment models include public cloud (shared infrastructure), private cloud (dedicated to one organization), and hybrid cloud (combining both).

Elaborate (8-10 minutes)

Examine how cloud computing works: virtualization, containerization, and load balancing distribute work across thousands of servers. Discuss elasticity—the ability to scale resources up or down automatically based on demand. Analyze cloud pricing models: pay-per-use, reserved instances, and spot instances. Real-world adoption: Netflix runs entirely on AWS, government agencies use Azure, and startups leverage free tiers to launch products.

Evaluate (5-8 minutes)

Students evaluate a scenario: a growing startup needs to decide between on-premise and cloud infrastructure. They must analyze cost, scalability, security, control, and maintenance for each option. Written recommendation with justification. Quiz: match service models to use cases, identify which cloud model (public/private/hybrid) suits different organizations.